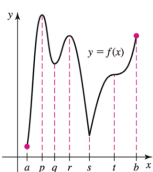


Problem 3

Problem 1 For each point in the interval (a,b) and identified on the graph below, determine if the function f has a critical point, a local max or min at that point.



Explanation. (p) The function f has a critical point and a local maximum at $x = p$.

(q) The function f has a critical point and a local minimum at $x = q$.

(r) The function f has a critical point and a local maximum at $x = r$.

(s) The derivative of f does not exist at $x = s$ because the function f has a corner at $x = s$. The function has a critical point and a local minimum at $x = s$.

(t) The function f has a critical point but not a local maximum or minimum at $x = t$.